Speech to Speech Translation Systems

Benchmarking and Implementation

Tanay Bhat (22B3303)
Jay Mehta (22B1281)
Kshitij Vaidya (22B1829)
Vatsal Melwani (22B0396)

IIT Bombay

May 1, 2026

Outline

Introduction

ASR: Overview

The ASR module of a S2S pipeline first converts the input speech into text. It involves several key components:

- **Audio Preprocessing:** Noise reduction and feature extraction using MFCCs, Mel spectrograms etc.
- **Encoder:** Responsible for capturing temporal dependencies and contextual information of the input audio using the extracted features. Typically implemented using RNNs, CNNs, or Transformers.
- **Decoder:** Generates the output text sequence based on the encoder reprensentation. Current state-of-the-art models often use attention mechanisms to focus on relevant parts of the input during decoding.

ASR Benchmarking: Setup

- Accent and dialect variation
- Background noise handling
- Real-time processing requirements
- Domain-specific vocabulary

ASR Benchmarking: Results

- Accent and dialect variation
- Background noise handling
- Real-time processing requirements
- Domain-specific vocabulary

ASR Benchmarking: Inferences

- Accent and dialect variation
- Background noise handling
- Real-time processing requirements
- Domain-specific vocabulary

MT: Overview

- Translates text from source to target language
- Evolution: Rule-based → Statistical → Neural
- Transformer architecture revolution
- Quality assessment and evaluation

MT: Key Approaches

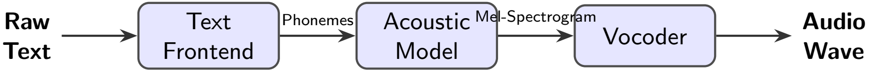
- Neural Machine Translation (NMT)
- Transfer learning and multilingual models
- Low-resource language handling
- Context-aware translation

What is Text-to-Speech (TTS)?

Text-to-Speech (TTS) is the artificial synthesis of human-sounding speech from raw text.

- **Core Objectives:**
  - *Intelligibility:* Clear, understandable pronunciation of words.
  - *Naturalness:* Accurate prosody, intonation, rhythm, and emotion.
- **Evolution of the Technology:**
  - **Concatenative (1990s-2000s):** Splicing together tiny fragments of pre-recorded human speech. Highly intelligible but sounded robotic and rigid.
  - **Parametric (2000s-2010s):** Statistical models (like HMMs) generating audio based on parameters. Smoother, but sounded muffled or "buzzy".
  - **Neural (Present):** Deep learning models predicting high-fidelity audio waves directly from text, achieving near-human naturalness.

The Standard Neural TTS Pipeline



- **Text Frontend:** Cleans and normalizes text (e.g., expanding ”180” to ”one hundred eighty”) and performs Grapheme-to-Phoneme (G2P) conversion to determine pronunciation.
- **Acoustic Model:** Maps phonemes to acoustic representations, predicting pitch, energy, and duration. *(Examples: Tacotron 2, FastSpeech)*
- **Vocoder:** Reconstructs the actual time-domain audio waveform from the highly compressed Mel-spectrogram. *(Examples: WaveNet, HiFi-GAN)*

Systems Under Evaluation

MMS-TTS

- **Architecture:** VITS-based (End-to-End).
- **Primary Strength:** Highly lightweight and efficient; state-of-the-art for low-resource languages.
- **Limitation:** Generally single-speaker per model with fixed prosody.

Parler-TTS

- **Architecture:** Autoregressive Transformer.
- **Primary Strength:** Granular control over emotion, tone, pacing, and background noise via text descriptions.
- **Limitation:** Higher inference latency due to autoregressive generation.

XTTS-v2

- **Architecture:** GPT-style + Mel/HiFi-GAN.
- **Primary Strength:** Requires only $\sim$3 seconds of reference audio; supports cross-lingual voice cloning.
- **Limitation:** High resource consumption compared to non-autoregressive models.

Evaluation Metrics for TTS Models

Intelligibility & Efficiency

- **Word Error Rate (WER)**
  What it is: Transcription accuracy of the audio using an ASR system. Lower is better.
- **Character Error Rate (CER)**
  What it is: Similar to WER, but captures finer phonetic/spelling errors. Lower is better.
- **Real-Time Factor (RTF)**
  What it is: Processing time divided by generated audio length. Lower is better.

Prosody & Expressiveness

- **Pitch Variance (F_0 Std Dev)**
  What it is: Range of intonation.
- **Energy Dynamics (RMS Std)**
  What it is: Variations in volume/loudness.
- **Speaking Rate**
  What it is: Speed (Syllables/words per sec).
- **Pause Ratio (Onset/s)**
  What it is: Frequency of breath/silence pauses.

TTS: Benchmarking Results

Model	Language	Latency (ms)	RTF	Throughput (cps)
MMS-TTS	English	2621.382	0.439	31.711
MMS-TTS	Hindi	2451.2891	0.432	27.49
Parler-TTS	English	68026.311	11.264	1.281
XTTS-v2	English	26182.9	3.438	3.491
XTTS-v2	Hindi	23687.88	3.204	2.718

Table: Benchmarking results for TTS models across Performance Metrics.

TTS: Benchmarking Results

Model	Language	WER	CER
MMS-TTS	English	34.892	17.968
MMS-TTS	Hindi	-	-
Parler-TTS	English	27.57	19.854
XTTS-v2	English	20.465	6.825
XTTS-v2	Hindi	-	-

Table: Benchmarking results for TTS models across Performance Metrics (Contd.).

**WER/CER for Hindi were turning out to be extremely high due to ASR model's poor performance on Hindi audio, so we have omitted those values to avoid skewing the analysis.*

TTS: Benchmarking Results

Model	Language	Pitch (Mean)	Pitch (Std)	Pitch Range
MMS-TTS	English	99.978	19.158	81.388
MMS-TTS	Hindi	148.216	37.359	176.392
Parler-TTS	English	233.055	54.52	253.809
XTTS-v2	English	141.029	29.674	136.078
XTTS-v2	Hindi	139.625	16.915	76.358

Table: Benchmarking results for TTS models across Prosodic Metrics (Contd.).

TTS: Benchmarking Results

Model	Language	Speaking Rate	Energy Dynamics	Pause Ratio
MMS-TTS	English	6.788	0.067	0.183
MMS-TTS	Hindi	5.808	0.093	0.246
Parler-TTS	English	5.35	0.082	0.153
XTTS-v2	English	4.284	0.099	0.212
XTTS-v2	Hindi	4.268	0.1002	0.186

Table: Benchmarking results for TTS models across Prosodic Metrics.

TTS: Plots

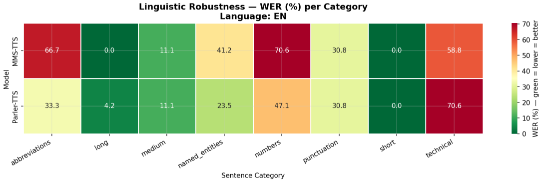


Figure: WER Heatmap for English for MMS-TTS and Parler-TTS for different types of sentences.

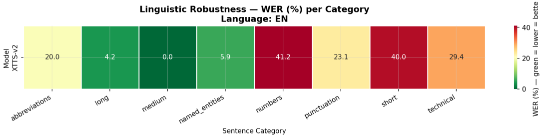


Figure: WER Heatmap for English for XTTS-v2 for different types of sentences.

End-to-End Pipeline

- ASR $\rightarrow$ MT $\rightarrow$ TTS workflow
- Use case: Real-time speech translation
- Latency and quality trade-offs
- Error propagation considerations

Pipeline Architecture

- Modular vs. end-to-end approaches
- API integration strategies
- Scalability and deployment
- Performance optimization

Applications

- International conference interpretation
- Language learning platforms
- Accessibility tools
- Customer service automation

Conclusion

- Integrated ASR-MT-TTS systems enable powerful applications
- Continued advances in neural architectures
- Challenges remain in low-resource scenarios
- Future directions: multimodal integration, personalization

Thank You

Thank you for your attention!

Questions?

Contact: your.email@example.com